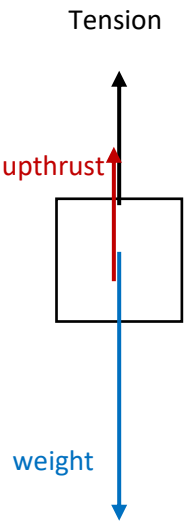
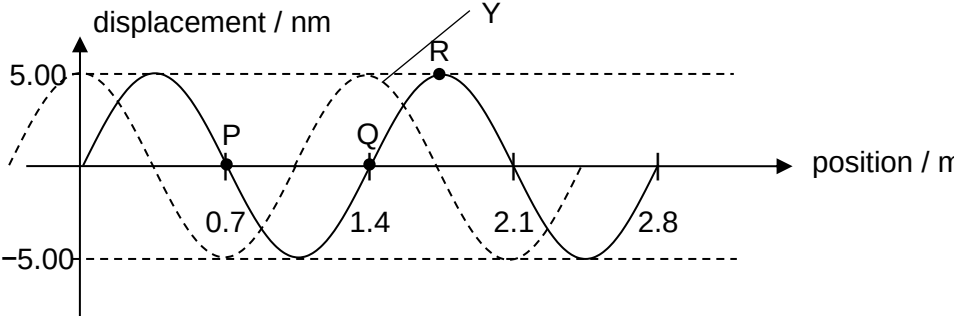
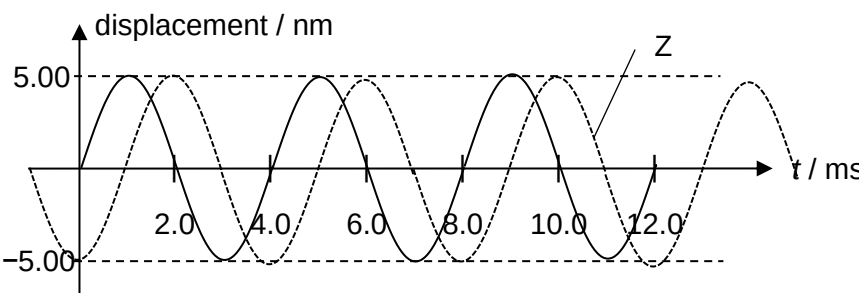


2022 Preliminary Examination Paper 2 Suggested Solution

Qn 1	Answer	Marks
(a) (i)	When mass M is placed on the spring, the spring extends by a length of $(L_2 - L_1)$. $mg = k(L_2 - L_1)$ $k = \frac{mg}{L_2 - L_1}$ $= \frac{(0.0985)(9.81)}{0.037 - 0.013} = 40.3 \text{ N m}^{-1}$	C1 M1
(a) (ii)	$\frac{\Delta k}{k} = \frac{\Delta m}{m} + \frac{\Delta(L_2 - L_1)}{L_2 - L_1} = \frac{0.2}{98.5} + \frac{0.2 + 0.1}{3.7 - 1.3} = 0.127$ Thus $\Delta k = k \times \frac{\Delta k}{k} = 40.3 \left(\frac{0.2}{98.5} + \frac{0.2 + 0.1}{3.7 - 1.3} \right) = 5 \text{ N m}^{-1}$ (Do not penalise if Δk is left to a value that is greater than 1 s.f. – this mark will be marked in (a)(iii))	M1 A1
(a)(iii)	$k = (40 \pm 5) \text{ N m}^{-1}$ Δk expressed to 1 s.f. and k expressed to the same place value as Δk. Allow ecf from (a)(ii) and (a)(iii).	A1
(b)(i)	 Marking Guidance: B1 – 3 forces drawn with correct direction B1 – Relative lengths of forces show $T+U = W$ [-1] – Missing labels/legend Note that first B1 mark must be obtained before the second B1 mark can be given. [Point of action is not marked here. But in general, students should know that the upthrust is acting at the centre of buoyancy – c.g. of the liquid displaced and weight is at the c.g. of the cube.]	B2
(b)(ii)	Initially (Fig. 1.2), P is inequilibrium, net force on it is zero. Weight = upthrust $W = (0.7V)\rho g$	M1
(b)(iii)	After cube Q was connected (Fig. 1.3), at new equilibrium, net force on P is zero. Weight of P = Tension due to string + new upthrust	

Qn 3	Answer	Marks
(a)	Using $v = f\lambda$ $v = \frac{1}{4.0 \times 10^{-3}} \times 1.4$ $= 350 \text{ m s}^{-1}$	M1 A1
(b)(i)	Particle R (As the particle is undergoing SHM, at the amplitude, the instantaneous velocity of the particle is zero).	A1
(b)(ii)	Particle Q . displacement / nm 5.00 -5.00 0.7 1.4 2.1 2.8 position / m Vertical lines represent equilibrium position of particle along the wave. Particle P is at the centre of compression and particle Q is at the centre of rarefaction	A1
(b)(iii)	Particle Q. (As the Fig. 3.2 shows a particle that is initially at equilibrium and is moving up in the next instant. Particle P and Q are both at equilibrium at $t = 0$. To determine which particle is presented above, the subsequent displacement-position graph in the next instant needs to be used. As seen below, since the wave is travelling towards the left, it is clear that particle P will be displaced in the negative direction and particle Q in the positive direction. Hence particle Q is depicted in Fig. 3.2). displacement / nm 5.00 -5.00 0.7 1.4 2.1 2.8 position / m Direction of wave progression is to the left.	A1
(c)(i)	Using equation . $\Delta\phi = \frac{\Delta x}{\lambda} \times 360^\circ = \frac{0.7}{1.4} \times 360^\circ$ $= 180^\circ$ Accept working with radians. And answer as π radians	M1 A1

(c)(ii)	Distance travelled by wave in 1 ms = $vt = (350)(1 \times 10^{-3}) = 0.35$ m The graph should have shifted to the left by 0.35 m.  Award mark as long as one full wavelength is drawn with displacement at 5.00 nm at initial position.	C1 A1
(c)(iii)	 At $t = 0$, Particle R is at +5.00 nm displacement so Particle S should be at -5.00 nm displacement (phase difference of π rad).	A1
	[Total : 10 marks]	